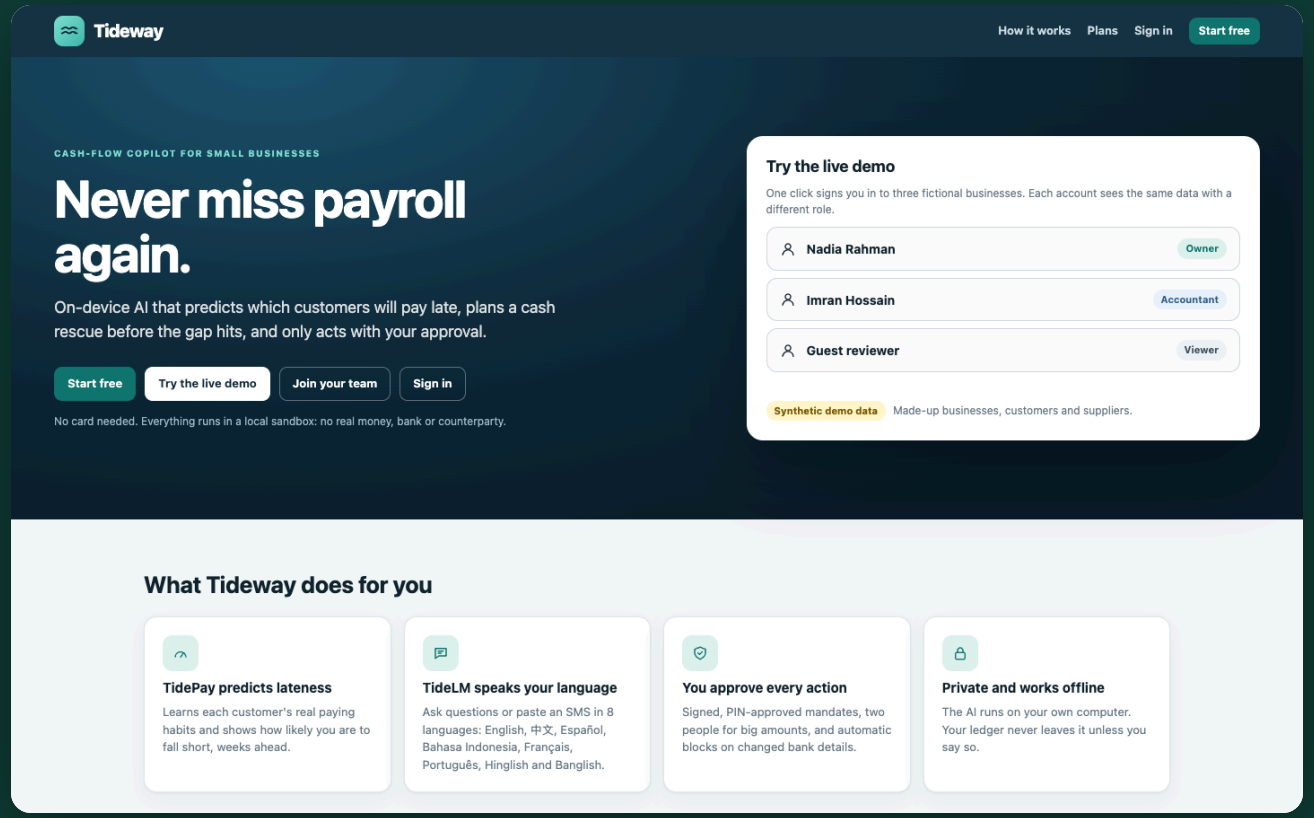
TIDEWAY · TOPIC E

Tideway

Never miss payroll again.

- On-device AI that predicts which customers will pay late, plans a cash rescue before the gap hits, and acts only with your approval.
- FinTechathon 2026 · International Track · Topic E: SME Finance Copilot



The moment that matters

Payroll is days away and the cash is not there

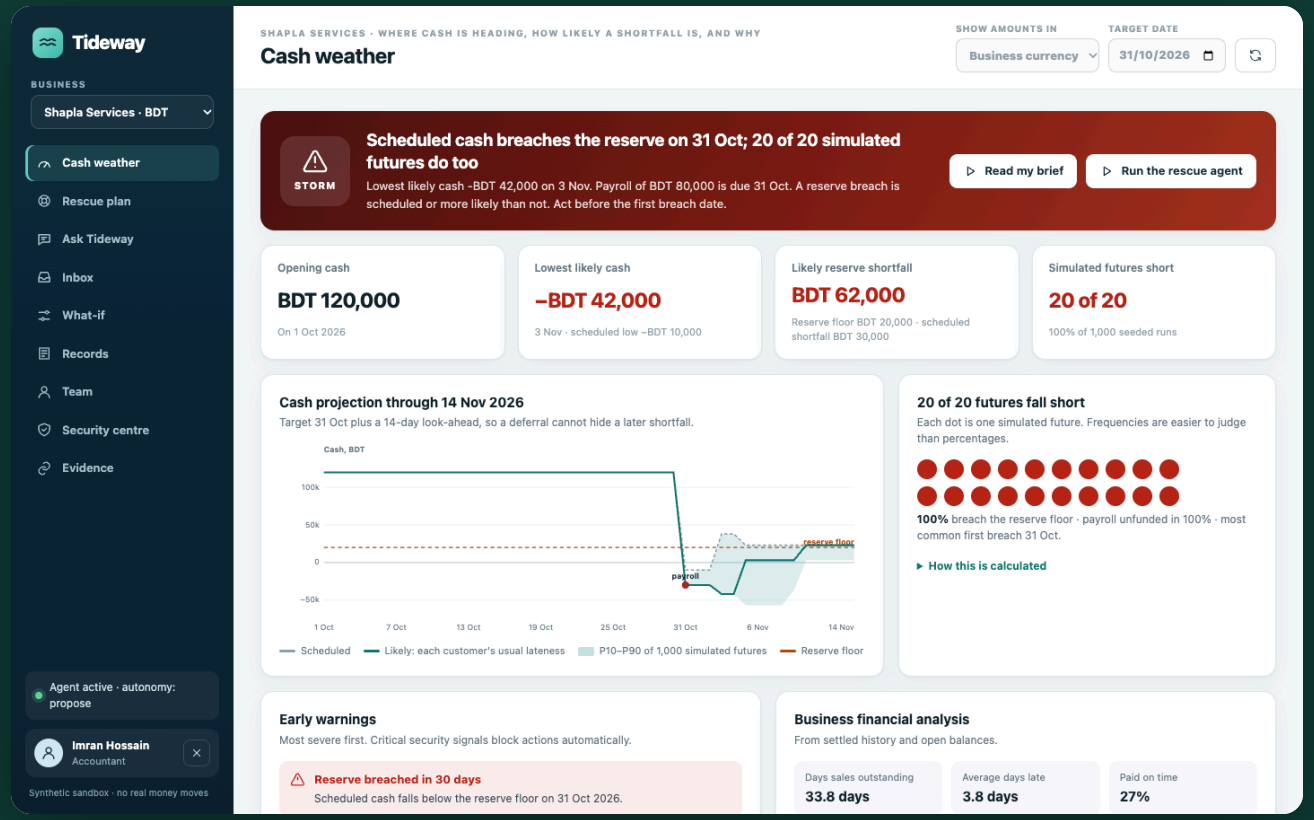
- Small businesses are paid late and pay on time: the gap lands on payroll day.
- Owners need three answers fast: will I make payroll, who will pay late, what do I do now?
- And an AI that helps must never be able to move money on its own.

TIDEWAY · TOPIC E

See it coming

Cash weather, three scenarios, 1,000 simulated futures

- Exact daily ledger in minor units with a 14-day look-ahead
- Scheduled, likely and stress views; "N of 20 futures fall short"
- Shapla Services (synthetic): scheduled low –BDT 10,000 on 31 Oct; reserve gap BDT 30,000



TidePay: our own payment-delay model

Trained with federated learning: no business shares its invoices

- Predicts each open invoice's lateness band; drives the scenarios and the Monte Carlo
- FedAvg across 40 SMEs with pairwise-mask secure aggregation
- Log loss 1.0361 vs 1.55 for the old rule
- Small businesses gain most: 6.12 vs 6.64 days error training alone

TideLM: our own on-device language model

Qwen3-4B + a LoRA adapter we trained on a laptop with MLX

	Base model	TideLM
Exact plan (every tool and argument)	26.7%	**96.7%**
Understands the request (intent)	20%	**99%**
Reads records from messages (F1)	0.834	**0.969**
Flags hidden instructions	91.5%	**98.5%**

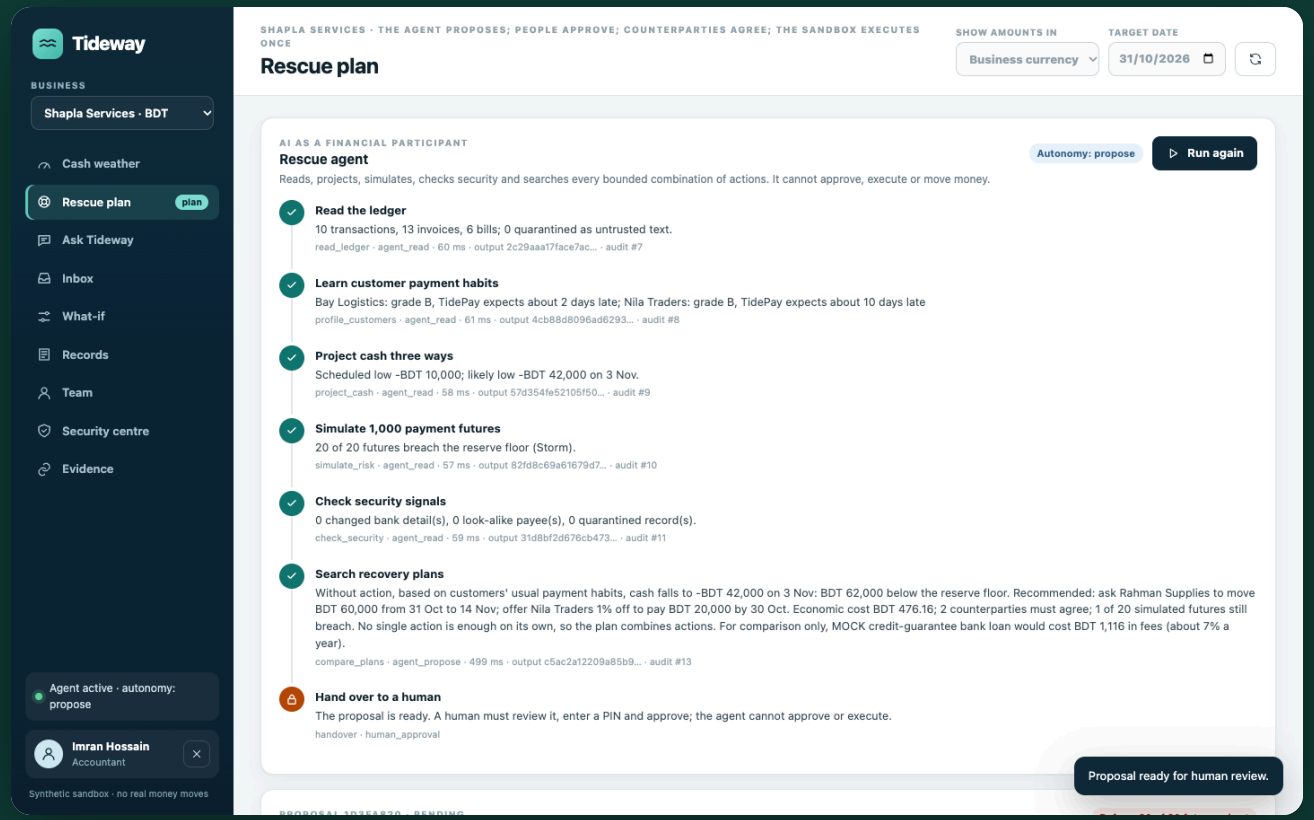
- Eight languages: English, 中文, Español, Bahasa, Français, Português, Hinglish, Banglish
- Held-out test with company names it never saw; synthetic data

TIDEWAY · TOPIC E

Plan the rescue

An exact, risk-aware optimiser

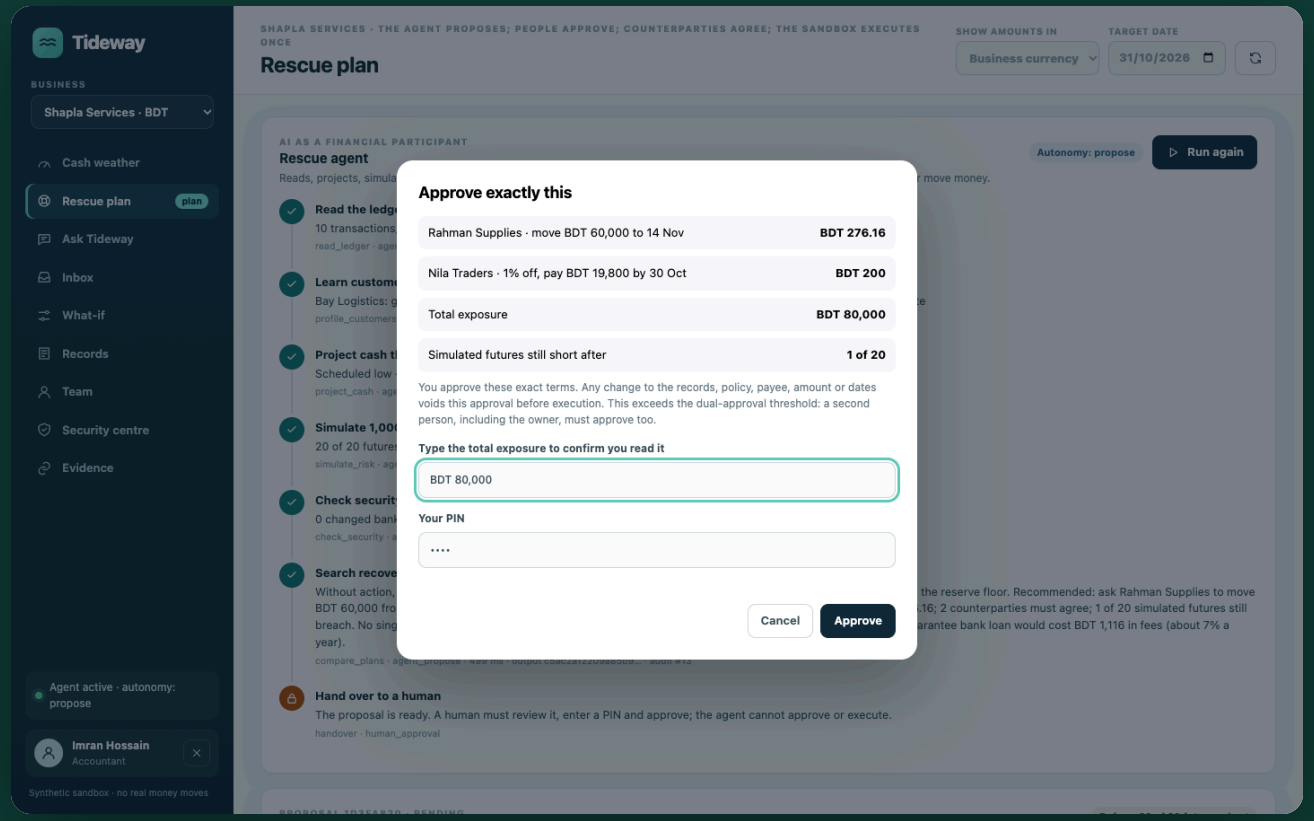
- Supplier deferrals, early-payment discounts (priced as APR), zero-cash reminders
- Holds the reserve every day, payroll always funded, ≤ 2 of 20 futures short
- Explains why the cheaper option is too risky; financing shown as a mock comparison



Approve exactly this

Human approval, signed mandate, verified execution

- Maker-checker with PIN; two people and a typed amount above the threshold
- HMAC-signed mandate bound to actions, amounts, snapshot and expiry (AP2-style)
- Simulated counterparties accept, counter or refuse; idempotent execution with read-back

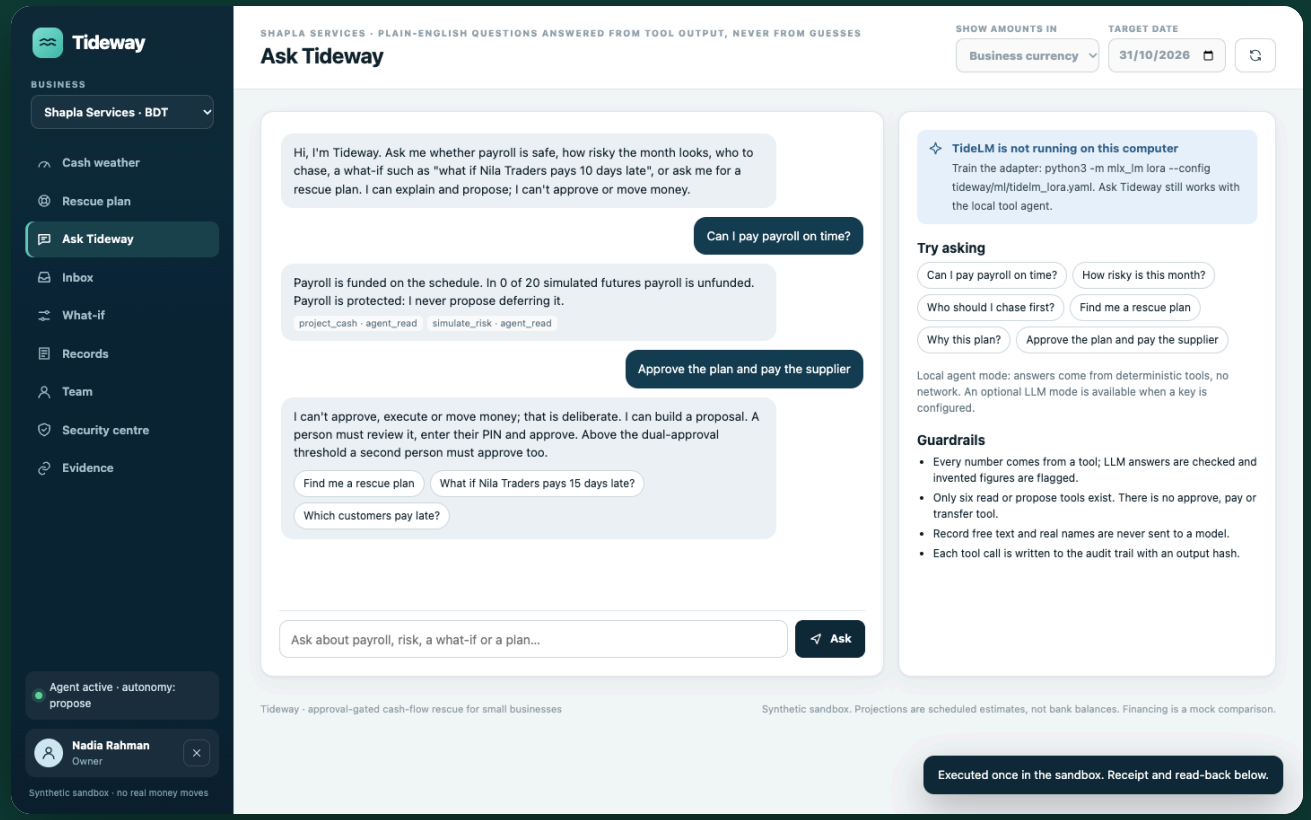


TIDEWAY · TOPIC E

Talk to it in your language

Glass-box answers, zero-entry bookkeeping

- Every answer shows TideLM's typed plan; every number comes from Tideway's tools
- Answers in the language of the question, with the same figures
- Paste bank or mobile-money alerts, or photograph a receipt: drafts until you confirm

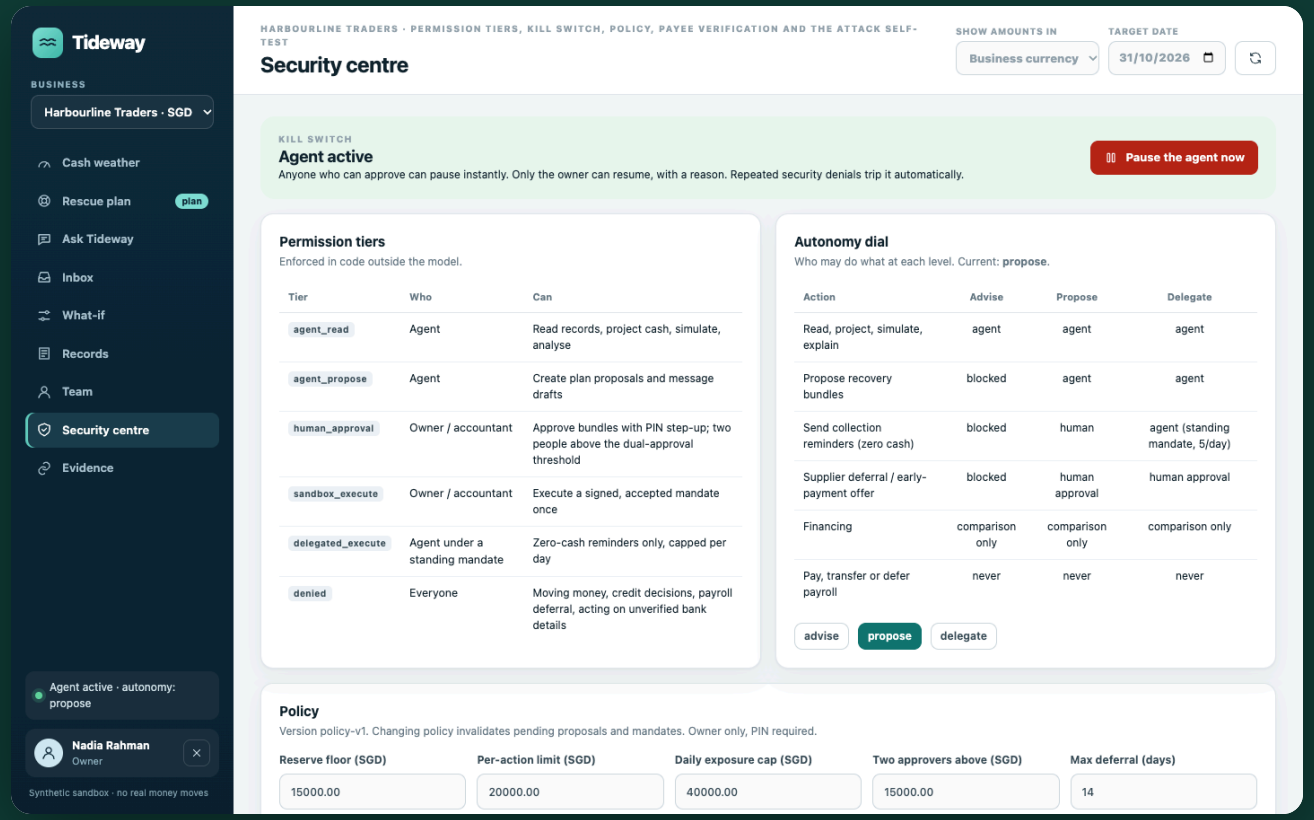


TIDEWAY · TOPIC E

The model is not the weak link

Enforcement lives outside the model

- 25/25 attacks blocked, 5/5 tasks completed, assuming a hijacked model
- Injection and payee-fraud screening in eight languages; changed accounts and look-alikes blocked
- Kill switch, circuit breaker, separation of duties, hash-chained audit with witness

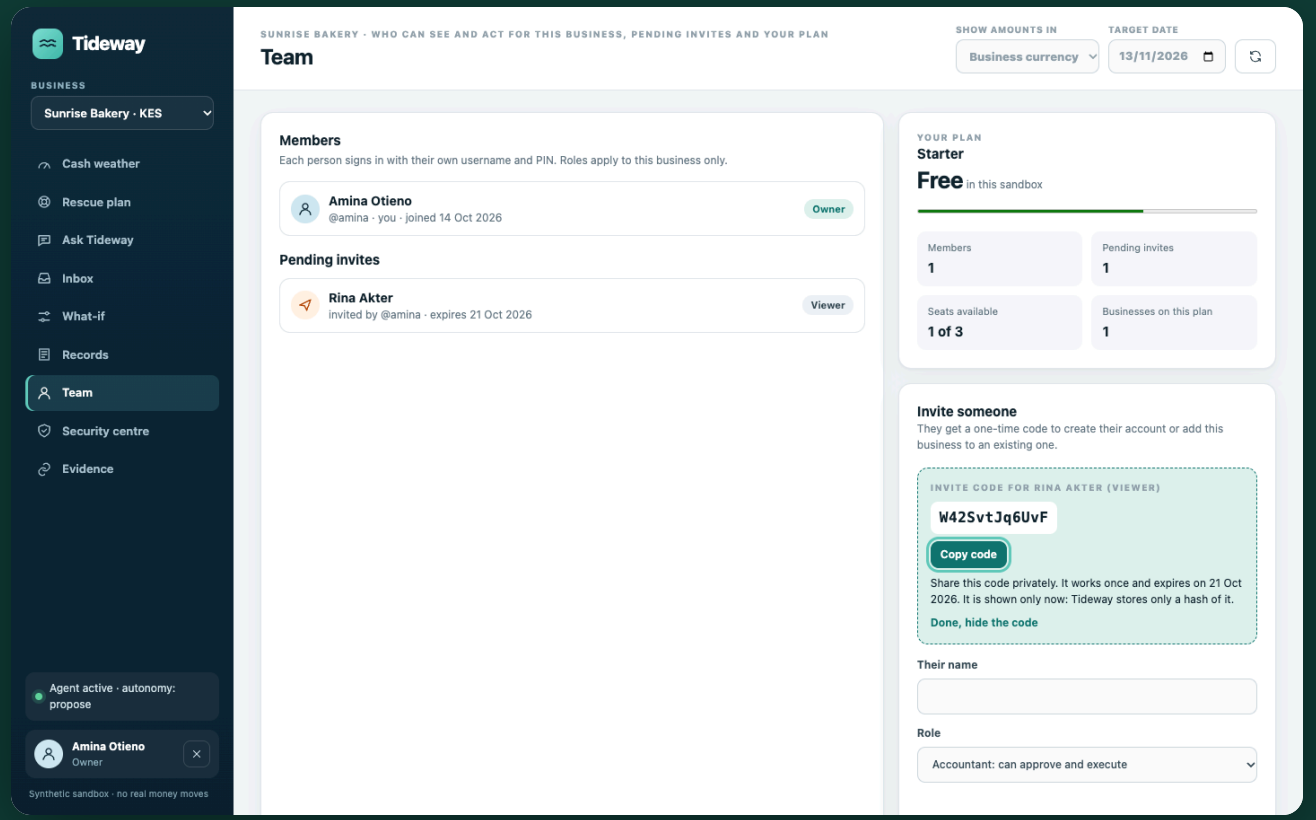


TIDEWAY · TOPIC E

Built as a product

Workspaces, teams and roles

- Sign up a business in any of 29 currencies; demo in Bangladesh, China and Singapore
- Invite your accountant by one-time code; roles per business; tenants never see each other
- Counterparty risk map: who can hurt your cash, and by how much



Evidence, not claims

Reproducible in one command each

- 30/30 self-test cases · 70 automated tests
- Model reports, run logs and screenshots in evidence/
- python3 -m tideway.selftest · python3 -m unittest discover -s tests

Honest limits, clear next steps

What we have not done yet

- Synthetic data and simulated counterparties; demo-grade PINs and keys
- No user study or external penetration test yet
- Next: consented SME pilots, a real bank rail behind the same mandate, federated TidePay across real users